

Voice Learning Knowledge Platform

Vision

Build a platform that transforms voice recordings into structured knowledge.

Users can:

- Record learnings directly from the browser
 - Upload audio recordings
 - Generate transcripts automatically
 - Create AI-powered summaries
 - Organize learnings into topics
 - Search past learnings
 - Share knowledge with teams
 - Build a collaborative knowledge base
-

Phase 1: Capture & Summarize (MVP)

Objective

Allow users to store learnings from voice recordings.

Features

Audio Input

- Record audio from browser
- Upload audio files

AI Processing

- Speech-to-text transcription
- AI-generated title
- AI-generated summary
- AI-generated category

Storage

- Save learnings in PostgreSQL

User Flow

Record / Upload

↓

Transcription

↓
Summary
↓
Category
↓
Store

Database

users

```
id
name
email
created_at
```

learnings

```
id
user_id
title
transcript
summary
category
created_at
```

Suggested Stack

- FastAPI
- PostgreSQL
- Whisper
- OpenAI / Gemini
- React / Next.js

Phase 2: Search Learnings

Objective

Allow users to find previous learnings quickly.

Features

- Full-text search
- Semantic search
- Topic filtering

- Date filtering
- Category filtering

Example Queries

What did I learn about GraphQL?

Show all FastAPI notes.

Find learnings related to Redis.

Infrastructure

Vector Database

- Qdrant

Embeddings

Store embeddings for:

- Transcript
- Summary

Phase 3: Smart Knowledge Organization

Objective

Automatically organize learnings.

Features

- Topic extraction
- Concept extraction
- Action item extraction
- Related learning detection

Example

Backend

- └─ FastAPI
- └─ GraphQL
- └─ PostgreSQL

Database

topics

```
id
name
description
```

learning_topics

```
learning_id
topic_id
```

Phase 4: Shared Workspaces

Objective

Enable collaborative learning.

Features

- Create workspaces
- Invite members
- Share learnings
- Team search

Example

```
Workspace:
System Design
```

```
Members:
User A
User B
User C
```

Database

workspaces

```
id
name
```

```
owner_id
created_at
```

workspace_members

```
workspace_id
user_id
role
```

workspace_learnings

```
workspace_id
learning_id
```

Phase 5: Collaborative Knowledge Base

Objective

Merge knowledge from multiple users.

Example

Input:

```
User A → GraphQL Queries
User B → GraphQL Resolvers
```

Output:

```
GraphQL
├── Queries
└── Resolvers
```

Features

- Duplicate detection
 - Topic merging
 - Combined summaries
 - Source attribution
-

Phase 6: Team Q&A (RAG)

Objective

Allow users to ask questions using stored knowledge.

Example

Explain GraphQL based on our team's learnings.

Flow

```
graph TD;
    Question --> VectorSearch[Vector Search];
    VectorSearch --> RelevantLearnings[Relevant Learnings];
    RelevantLearnings --> LLM;
    LLM --> Answer;
```

Features

- Workspace chatbot
 - Source citations
 - Related learning suggestions
-

Phase 7: Live Learning Mode

Objective

Process learnings in real time.

Features

- Live transcription
- Live summary updates
- Topic detection while speaking

Flow

```
Microphone
  ↓
WebSocket
  ↓
Realtime Transcript
  ↓
Realtime Summary
```

Phase 8: Learning Intelligence

Objective

Act as a learning assistant.

Features

Revision Reminders

You learned Binary Search 30 days ago.

Knowledge Gaps

You learned GraphQL but not Authentication.

Related Concepts

FastAPI is related to your GraphQL notes.

Progress Tracking

- Learning streaks
 - Topic mastery
 - Revision history
-

Phase 9: Knowledge Graph

Objective

Build a connected knowledge network.

Example

```
Backend
|
FastAPI
|
GraphQL
|
Resolvers
```

Features

- Visual graph view
 - Topic relationships
 - Concept navigation
-

Phase 10: AI Learning Coach

Objective

Provide personalized learning assistance.

Features

Study Plans

```
Create a GraphQL study plan.
```

Interview Preparation

```
Generate interview questions from backend learnings.
```

Quiz Generation

```
Test me on FastAPI.
```


Flashcards

Automatic flashcard generation from learnings.

Recommended Initial Scope

For Version 1:

- Phase 1
- Phase 2
- Phase 3
- Phase 4
- Phase 6

This provides:

- Audio Processing
- AI Summarization
- Semantic Search
- RAG
- Collaboration
- FastAPI Backend
- PostgreSQL
- Qdrant
- LLM Integration

while remaining achievable for a portfolio project.